Wireless Power by Resonant Magnetic Coupling

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2 Executive Summary

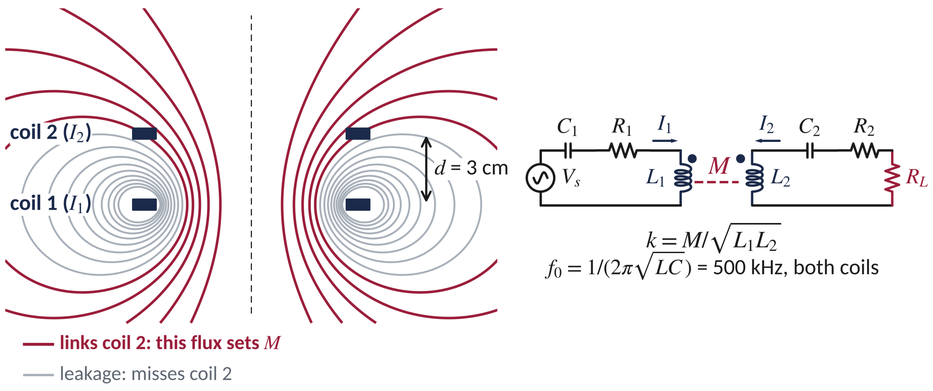
Two identical 10-turn, 10 cm air-core spiral coils, series-tuned to 500 kHz, transfer power across a 1–10 cm gap with no wire between them. At 500 kHz the wavelength is 600 m, so the coils interact through the reactive near field: primary current creates a magnetic field (Ampère), part of its flux links the secondary (mutual inductance M), and Faraday's law induces the secondary voltage. Resonance cancels the coil reactance, so performance is set by coupling k and quality factor $Q = 118.3$. A Neumann-integral model gives k falling from 0.51 at 1 cm to 0.030 at 10 cm. A phasor circuit model, confirmed by closed-form theory and LTspice, predicts maximum efficiencies of 97 % at 1 cm, 86 % at 5 cm and 57 % at 10 cm. With the 6 Ω load (optimal at 5 cm), delivered power peaks at critical coupling (6.4 W near 9.75 cm), not where efficiency is highest. All results are analytical and simulated.

3 The Physical Concept

Why it matters. With no plugs or contacts, a device can be sealed, waterproof, or even inside the body, so wireless power now reaches from phones to electric vehicles and deep-sea robots.

How it works. An alternating current in coil 1 makes a magnetic field. Part of that field passes through coil 2 and, because it changes, induces a voltage there (Faraday's law). A capacitor in series with each coil makes it resonate: energy sloshes between the coil's magnetic field and the capacitor's electric field, so a small drive builds a large current. Both coils are tuned to the same f_0, so power crosses the gap to R_L.

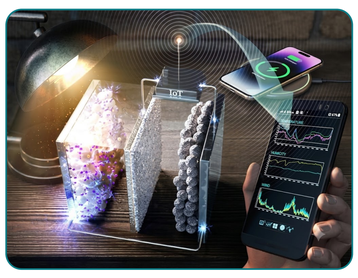
Why it is short range. The field is stored near the coils rather than radiated, so k falls from 0.51 at 1 cm to 0.03 at 10 cm.



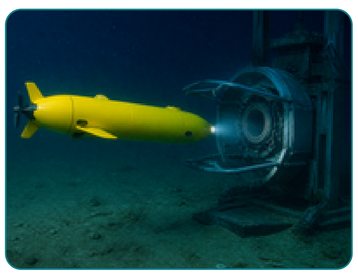
Left: field lines of coil 1, computed for our coils at d = 3 cm; wine lines thread coil 2 and set M. Right: the series-series circuit. Photos below: AI-generated illustrations.



Electric vehicles: a ground pad charges a parked robotaxi.



Phones and IoT: pads charge phones and power sensor nodes.



Underwater robots: AUVs recharge at seabed docks.

4 Governing EM Physics

(1) $\nabla \times \boldsymbol{H} = \boldsymbol{J} + \partial \boldsymbol{D}/\partial t \approx \boldsymbol{J} \qquad (k_0 a = \omega a/c \approx 5 \times 10^{-4})$

Ampère, quasi-static: the coil is 1/6000 of λ = 600 m, so displacement current is negligible.

(2) $\oint \boldsymbol{E}\cdot \mathrm{d}\boldsymbol{l} = -\mathrm{d}\Phi/\mathrm{d}t \;\Rightarrow\; V_2 = -j\omega M I_1$

Faraday: the changing shared flux induces the EMF that drives coil 2.

(3) $B_z = \mu_0 N I a^2 \,/\, [2(a^2 + z^2)^{3/2}]$

$M = \Phi_{21}/I_1 = (\mu_0/4\pi) \oint\oint \mathrm{d}\boldsymbol{l}_1 \cdot \mathrm{d}\boldsymbol{l}_2 \,/\, r$

Biot–Savart, Neumann: flux of coil 1 through coil 2 per amp. Pure geometry; $M_{12} = M_{21}$.

(4) $k = M/\sqrt{(L_1 L_2)}, \qquad k \propto 1/d^3 \;\text{ for }\; d \gg a$

Coupling: ours is 0.513, 0.111, 0.030 at 1, 5, 10 cm; $1/d^3$ only for $d \gtrsim 4a$.

(5) $f_0 = 1/(2\pi\sqrt{(LC)}), \qquad Q = \omega_0 L/R, \qquad f_\pm = f_0/\sqrt{(1 \pm k)}$

Resonance: 17.13 µH, 5.915 nF → 500 kHz; Q = 118.3 at 0.455 Ω. Strong k splits the peak.

(6) $Z_1 I_1 + j\omega M I_2 = V_s, \qquad j\omega M I_1 + Z_2 I_2 = 0$

$Z_{in} = R_1 + \omega_0^2 M^2/(R_2 + R_L) \quad \text{at } f_0$

Circuit, Poynting: at f_0, $W_m = W_e$ in each coil, so the source sees only resistance.

(7) $\eta_{max} = x \,/\, (1 + \sqrt{(1 + x)})^2, \qquad x = k^2 Q_1 Q_2$

Efficiency: at $R_L = R_2\sqrt{(1 + x)}$. At 5 cm x = 173, η_{max} = 85.9 %, R_L = 6.0 Ω.

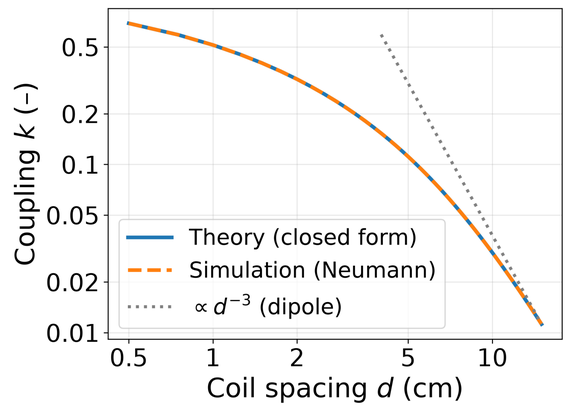
(8) $\delta = \sqrt{(2/(\omega\mu_0\sigma))} = 0.094$ mm (Cu, 500 kHz)

Skin depth: current crowds into a skin thinner than the 0.35 mm wire radius, raising R.

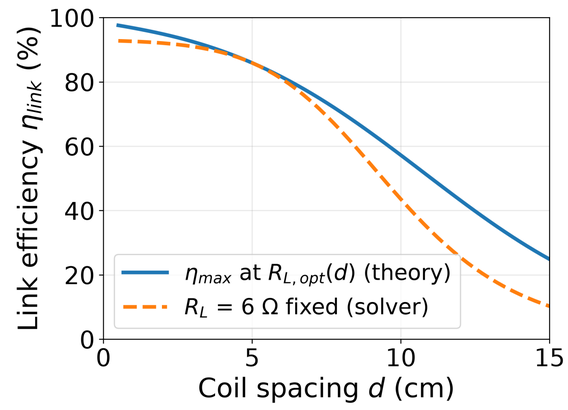
Boundary conditions: $\hat{n}\cdot(\boldsymbol{B}_1 - \boldsymbol{B}_2) = 0$, $\hat{n} \times (\boldsymbol{H}_1 - \boldsymbol{H}_2) = \boldsymbol{K}$. Metal forces $B_n \approx 0$ (eddy currents): L, k fall and an Al plate 10 mm away adds 87 mΩ. Ferrite guides flux, raising k.

Symbols: N turns, a coil radius, d spacing; I_1, I_2 into the dots; $Z_i = R_i + j\omega L_i + 1/(j\omega C_i)$, R_L added in Z_2; σ conductivity; $\mu_0 = 4\pi \times 10^{-7}$ H/m.

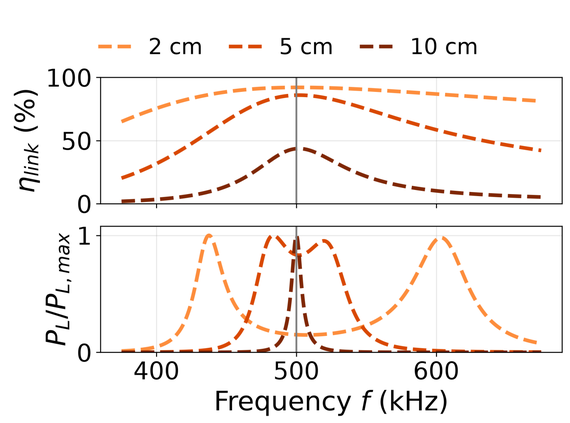
5 Analysis and Simulation



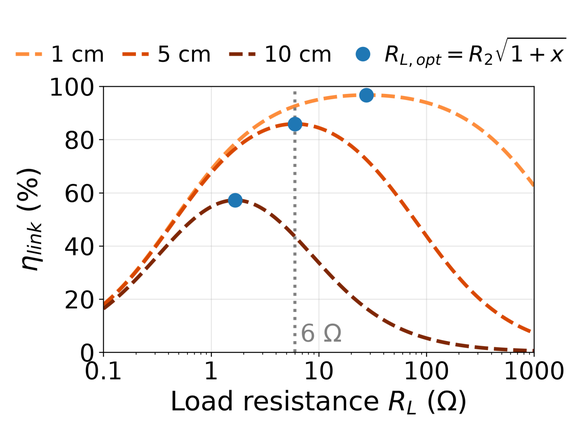
Closed form and Neumann agree to ≤ 0.05 % (0.5–15 cm); $k \propto 1/d^3$ only for $d \gg a$.



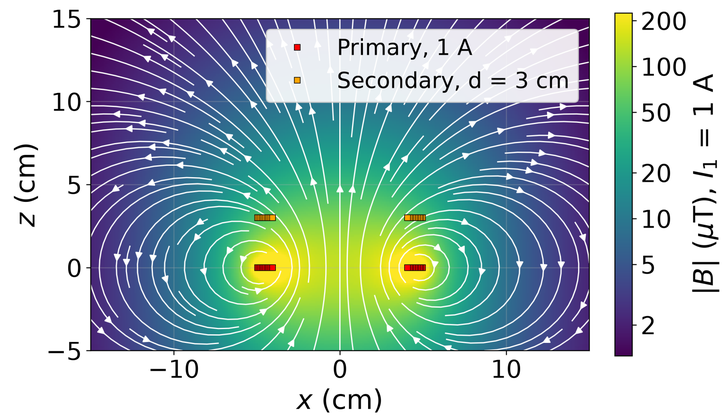
η_{max} is 97 % at 1 cm, 86 % at 5 cm, 57 % at 10 cm; the fixed 6 Ω load gives 44 % at 10 cm.



At 2 and 5 cm the power peak splits in two; at 10 cm a single sharp peak sits at f_0.



The best load $R_{L,opt} = R_2\sqrt{(1+x)}$ moves with spacing; 6 Ω is best only at 5 cm.



$|B|$ (µT) in the x–z plane: primary at 1 A, coil 2 at d = 3 cm.

Setup

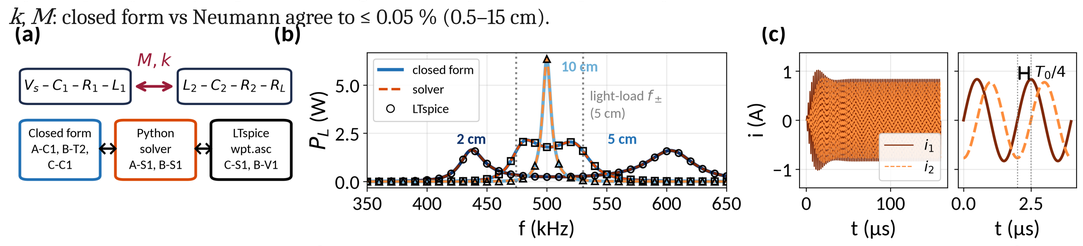
Planar spiral, N = 10
d_{out} = 100 mm
wire 0.7 mm, pitch 1.0 mm
L = 17.13 µH, C = 5.915 nF
f_0 = 500 kHz
R = 0.455 Ω, Q = 118.3
R_L = 6.0 Ω
V_s = 5 V peak, R_s = 0

6 Demonstration: Model vs Model

- **Validation chain:** closed form ↔ Python Neumann sum + 2 × 2 phasor solver ↔ LTspice sim/wpt.asc. LTspice agrees with the solver to 5×10^{-10} % (2, 5, 10 cm, 350–650 kHz).
- Faraday in time: at f_0, $V_2 = -j\omega M I_1$ and $Z_2 = R_2 + R_L$ is real, so i_2 lags i_1 by 90° ($T_0/4$ = 0.50 µs).
- Light-load peaks $f_\pm$ (eq. 5) give k within 0.3 % at 1, 2, 5 cm (5 cm: 474.4 / 530.3 kHz → 0.1109).
- **Finding:** a 6 Ω load pulls them in (5 cm: 483.5 / 519.6 kHz → k = 0.072, 35 % low; 10 cm: 1 peak).

d (cm)	M (µH)	k	ηmax (RL,opt)	η at 6 Ω	PL at 6 Ω (W)	Regime
1	8.79	0.513	96.8 % (27.6 Ω)	92.6 %	0.098	over-coupled
3	3.72	0.217	92.5 % (11.7 Ω)	91.0 %	0.526	over-coupled
5	1.90	0.111	85.9 % (6.00 Ω)	85.9 %	1.79	over-coupled; design point
10	0.51	0.030	57.2 % (1.67 Ω)	43.5 %	6.36	near critical (k < kc)

V_s = 5 V pk, R_s = 0, $R_1 = R_2$ = 0.455 Ω; regime from k_c = 0.0318 (R_L = 6 Ω, R_s = 0).
k, M: closed form vs Neumann agree to ≤ 0.05 % (0.5–15 cm).



(a) Model chain. (b) P_L (W) vs f(kHz) at 2, 5, 10 cm, R_L = 6 Ω; markers: LTspice; dotted lines: light-load $f_\pm$ at 5 cm. (c) i_1, i_2 (A) vs t (µs), 5 cm: i_2 lags i_1 by 90°. Read k only from light-load peaks.

7 Conclusion and References

Key findings

- Q = 118.3: η_{max} = 96.8 % at 1 cm, 85.9 % at 5 cm ($R_{L,opt}$ = 6.0 Ω), 57.2 % at 10 cm.
- Max $\eta \neq$ max power: at 5 cm, $R_{L,opt}$ = 6.0 Ω, but R_L = 79.1 Ω gives the most power.

Limitations

- Range: $M \propto 1/d^3$ only for $d \gg a$ (log-log slope −2.1 at 8–10 cm, −2.97 at 50–100 cm).
- AC loss: R_{dc} 0.13 → 0.455 Ω (skin 0.269, prox. 0.136, ESR 0.035, joints 0.015); range 0.38–0.48 Ω.
- Misalignment (3 cm): η = 91.1 / 88.8 / 84.2 % at Δx = 2 / 3 / 4 cm; M = 0 at 8.22 cm, then M < 0.
- Al plate 5 mm behind coil 2: k −56.4 %, R_2 +178 mΩ, L_2 −51 %; ferrite: k +34.5 %. η (3 cm): 92.5 → 24.0 % (63.9 % with C_2 retuned).
- Stress (R_s = 0): V_{C1} 44.8 / 314 V pk at 5 / 10 cm; $I_1 \leq$ 11.0 A pk, 27.5 W (50 Ω source: 0.1 A). Field: 82.3 µT rms at z = 0, < 27 µT rms (ICNIRP 2010) beyond 4.7 cm, < 1.84 beyond 15.4 cm.

Future work: build and measure a prototype; ferrite backing; adaptive frequency tuning [2].

[1] A. Kurs et al., Science, vol. 317, no. 5834, pp. 83–86, 2007.
[2] A. P. Sample, D. A. Meyer, and J. R. Smith, IEEE Trans. Ind. Electron., vol. 58, no. 2, pp. 544–554, 2011.
[3] S. Y. R. Hui, W. Zhong, and C. K. Lee, IEEE Trans. Power Electron., vol. 29, no. 9, pp. 4500–4511, 2014.
[4] R. K. Shevgaonkar, Electromagnetic Waves. New Delhi: Tata McGraw-Hill, 2006.